Setup Guide A

Your Own Laptop

MLOps & Model Deployment

Vel Tech Rangarajan Dr. Sagunthala R&D Institute

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Dr. Venkateshwaran (Venkat) Loganathan

Senior Director of Engineering, ConverSight.ai

Setup Guide A — Your Own Laptop

Do this before Day 1. Budget **45 minutes**, and do it on the network you'll actually use. If you get stuck, don't panic: **Setup Guide B (Codespaces)** needs nothing installed and works in a browser. You can switch at any point, even mid-workshop.

⚠ Read this first — the one thing that breaks most laptops

> **Your Python must be 3.10, 3.11 or 3.12. Python 3.13 or newer will NOT work.**
>
> Our pinned `numpy==1.26.4` has no build for Python 3.13+. If you install the newest Python from python.org, you will get this error and nothing will work:
>
> `ERROR: Could not find a version that satisfies the requirement numpy==1.26.4 ERROR: No matching distribution found for numpy==1.26.4`
>
> **Python 3.11 is the tested, recommended version.** Check yours with `python --version` *before* you install anything else.

What you need to install

#	Software	Version	Needed for	Download
1	**Python**	**3.11** (3.10 / 3.12 fine · **not 3.13+**)	Everything	python.org/downloads
2	**Git**	any recent	Getting the project, Labs 4 & 5	git-scm.com/downloads
3	**Docker Desktop**	any recent	**Labs 3 & 5 only**	docker.com/products/docker-desktop
4	**VS Code**	any recent	Recommended editor	code.visualstudio.com

Also create two **free** accounts and **verify the confirmation email** for each:

- **GitHub** — github.com (Labs 4 & 5)
- **Hugging Face** — huggingface.co (Lab 5)

Disk space: ~10 GB free (Docker Desktop is most of it). **RAM:** 8 GB or more.

Step 1 — Install Python

Windows

1. Go to **python.org/downloads/windows** and pick **Python 3.11.x** → *Windows installer (64-bit)*. Do **not** just click the big yellow button on the homepage — that gives you the newest version.

2. Run the installer. **Tick ☑ "Add python.exe to PATH"** on the first screen. This is the single most common Windows mistake.
3. Click *Install Now*.
4. Open a **new** PowerShell window and check:

```
python --version
```

You want `Python 3.11.x`. If it prints nothing, opens the Microsoft Store, or says `'python' is not recognized`, PATH wasn't set — re-run the installer, choose *Modify*, and tick the PATH box.

macOS

```
# with Homebrew (recommended)
brew install python@3.11
python3.11 --version
```

No Homebrew? Install the **3.11.x** macOS 64-bit universal2 installer from python.org.

> On macOS, use **`python3`** (or `python3.11`), not `python`. Plain `python` may not exist.

Linux (Ubuntu/Debian)

```
sudo apt update
sudo apt install -y python3.11 python3.11-venv python3-pip git
python3.11 --version
```

Step 2 — Install Git

- **Windows:** git-scm.com/downloads → run the installer → accept the defaults.
- **macOS:** `brew install git` (or run `git --version`, which prompts to install Xcode tools).
- **Linux:** included in Step 1 above.

Verify:

```
git --version
```

Step 3 — Install Docker Desktop *(needed for Labs 3 & 5 — Day 2 onward)*

You can leave this until the evening of Day 1, but **don't leave it until Day 2 morning** — it needs a restart and sometimes a BIOS change.

1. Download **Docker Desktop** for your OS and install it.
2. **Windows only:** it will ask to enable **WSL 2**. Say yes, and **reboot** when asked. If Docker later complains that virtualization is disabled, you must enable **Virtualization / SVM / VT-x** in your BIOS — search your laptop model plus "enable virtualization in BIOS".

3. **Launch Docker Desktop** and wait until the whale icon says **"Docker Desktop is running"**. Docker only works while this app is open.
4. Verify:

```
docker run hello-world
```

You want to see **"Hello from Docker!"**. If you get `Cannot connect to the Docker daemon`, Docker Desktop isn't running — open the app and wait.

Step 4 — Get the sample project

You'll do everything in this one repository.

Option 1 — Git clone *(recommended: Labs 4 & 5 need git anyway)*

```
cd ~/Desktop
git clone https://github.com/anandvenkat4/mlops-sample-project
cd mlops-sample-project
```

Option 2 — Download the ZIP *(no git needed)*

1. Open **github.com/anandvenkat4/mlops-sample-project**
2. Green **Code ▾** button → **Download ZIP**
3. **Unzip it** (don't work inside the zip preview), then `cd` into the folder.

> **Windows tip:** put the folder somewhere short and simple like `C:\mlops`, **not** inside OneDrive/Desktop with spaces in the path. OneDrive sync corrupts virtual environments.

You should now see these files:

```
app/  src/  tests/  solutions/  .github/  .devcontainer/
Dockerfile  requirements.txt  train.py  make_data.py  README.md
```

Step 5 — Create a virtual environment

A venv keeps this project's packages separate from the rest of your machine.

Windows (PowerShell):

```
python -m venv .venv
.venv\Scripts\activate
```

macOS / Linux:

```
python3.11 -m venv .venv
source .venv/bin/activate
```

Your prompt should now start with `(.venv)`. That means it worked.

> **Windows error** "*running scripts is disabled on this system*"? Run this once, then activate again: `powershell Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

⚠ **You must re-run the activate command every time you open a new terminal.** If a command suddenly says "module not found", you almost certainly forgot to activate.

Step 6 — Install the dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
```

This downloads roughly 300 MB and takes 3–10 minutes. Grab a coffee.

Step 7 — Verify everything works

Run all four. This is exactly what the labs will need:

```
python make_data.py       # 1. creates data/loan.csv
python train.py           # 2. trains + logs to MLflow
pytest -q                 # 3. tests pass
docker run hello-world    # 4. Docker works (Labs 3 & 5)
```

✓ Your "I'm ready" checklist

Check	What you should see
`python --version`	`3.10`, `3.11` or `3.12` — **not 3.13+**
Prompt shows `(.venv)`	Virtual environment is active
`python make_data.py`	`data/loan.csv` now exists
`python train.py`	prints `accuracy`, `f1`, `roc_auc` and creates `model.joblib`
`pytest -q`	all tests pass, no red
`docker run hello-world`	**"Hello from Docker!"**
GitHub account	created **and email verified**
Hugging Face account	created **and email verified**

If all eight pass, you are fully ready for all three days.

Troubleshooting — the errors you're most likely to hit

Error / symptom	Cause	Fix
`No matching distribution found for numpy==1.26.4`	**Python 3.13+**	Install Python **3.11** and rebuild the venv (delete `.venv`, redo Steps 5–6)
`'python' is not recognized` (Windows)	PATH not set	Re-run the Python installer → *Modify* → tick **Add to PATH**; open a **new** terminal
`python: command not found` (macOS)	macOS uses `python3`	Use `python3` or `python3.11`
`running scripts is disabled` (Windows)	PowerShell policy	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`
`ModuleNotFoundError: No module named 'sklearn'`	venv not activated, or installed outside it	Activate the venv (prompt shows `(.venv)`), then re-run `pip install -r requirements.txt`
`Cannot connect to the Docker daemon`	Docker Desktop not running	Open Docker Desktop, wait for "running"
Docker: *virtualization not enabled* (Windows)	VT-x/SVM off in BIOS	Enable virtualization in BIOS; enable WSL 2
`pip` is extremely slow / times out	Campus network filtering	Tether to your phone, or switch to **Codespaces**
`port 5000 already in use` (macOS)	AirPlay Receiver uses 5000	System Settings → General → AirDrop & Handoff → turn **off** AirPlay Receiver, or use `--port 5001`
Everything breaks after moving the folder	venv stores absolute paths	Delete `.venv` and redo Steps 5–6

Which labs need what

Lab	Day	Python	Docker	GitHub	Hugging Face
1 · MLflow tracking	1	✓	—	—	—
2 · FastAPI serving	2	✓	—	—	—
3 · Docker	2	✓	✓	—	—
4 · CI/CD	3	✓	—	✓	—
5 · Cloud deploy	3	✓	✓	✓	✓
6 · Monitoring & drift	3	✓	—	—	—

So if Docker defeats you, you can still do Days 1 and most of Day 2 — but sort it out before Day 3.

Reading the lab worksheets on a laptop

Every lab shows **two** command blocks. Use the one labelled **"On your own laptop"** and ignore the Codespaces block. In the laptop path:

- your project folder is wherever you cloned it (e.g. `~/Desktop/mlops-sample-project`)
- you **do** run `source .venv/bin/activate` (Windows: `.venv\Scripts\activate`)
- `http://127.0.0.1:PORT` in your browser works exactly as printed

If your laptop fights back on the day

Switch to Codespaces and keep going — see Setup Guide B. It takes about 3 minutes, needs only a browser, and your work is identical. Nobody should lose lab time to an install problem. Tell the instructor you've switched so the demo instructions match your screen.